

Comparison of different CNN Architecture for Binary Classification of Retinal Diseases

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Abstract—The most common causes of vision impairment include retinal conditions like diabetic retinopathy, glaucoma and age-related macular degeneration, which is why they have to be detected promptly and precisely. Conventional diagnostic methods are based on manual examination of fundus photographic images by ophthalmologists, which is time-consuming, subjective and challenging to scale [1]. The paper provides a comparative evaluation of Convolutional Neural Network (CNN) models to do binary classification of retinal diseases on the RFMiD dataset [6]. The suggested model compares a base CNN, transfer learning model founded on ResNet50 [13], and a model that uses preprocessing to improve the edge detector, the Sobel filter [7]. Evaluation of performance is done based on accuracy, precision, recall, specificity, and F1-score [14], [15]. Findings indicate that the models trained with raw images are underfitted, and the Sobel-based preprocessing would increase the accuracy but decrease the sensitivity. Custom CNN has a high recall, but lacks specificity because of the class imbalance. This paper identifies the necessity to design the model in a balanced way, to preprocess the data efficiently, and to handle the imbalance in the data in order to obtain reliable automated diagnosis.

I. INTRODUCTION

A. Effect on Life / Society

Permanent blindness and visual impairment are currently a global health issue of concern mainly caused by Retinal Diseases. A wide range of retinal diseases, such as diabetic retinopathy, glaucoma, and age-related macular degeneration affects millions of people every year. The rise of these diseases are due to aging trends, unhealthy lifestyles, and the increased number of chronic diseases. Vision impairment reduces quality of life and makes everyday tasks more difficult for a person. At worst, it may result in total blindness, triggering emotional, social and economic costs to the patients and their families. Most of the retinal diseases are either treatable or preventable when diagnosed at an early age. Hence, there is an immediate requirement of effective, precise and scalable screening systems that could help medical practitioners to diagnose retinal illnesses and alleviate the general strain on health care systems.

B. Traditional Method their drawbacks

Retinal diseases are conventionally diagnosed by ophthalmologists by the manual analysis of retinal fundus images, by means of the ophthalmoscopy technique and fundus photography. The most popular and common method is medical

experts scrutinize the images keenly to identify abnormalities, including lesions, haemorrhages, exudates, and structural changes in the retina. Although this method is clinically valid, it poses a number of limitations intrinsically. To begin with, it is a time consuming process that needs highly trained experts making it inappropriate when handling large scale screening programs. Secondly, because experts differ in experience and interpretation, diagnosis can be different between experts, thus creating inter-observer variability. Also, in most places especially the rural and underdeveloped ones, there is a lack of qualified ophthalmologists hence getting timely diagnosis is not easy. With the increasing population of patients, manual screening becomes more ineffective and inefficient thereby leading to late and improper diagnosis and development of retinal diseases. To address these difficulties and automated solutions is necessary to help in quicker and more reliable diagnosis in modern society

C. Significant Limitations and Objective of the research

Automated systems have been developed based on machine and deep learning methods, although there are still a number of challenges in existing methods. Most methods are based on multi-class classification or detecting a particular disease, complicating the models and raising computational demands. Other challenges exist like the imbalance of classes, overfitting, and ineffective generalisation of the systems with different datasets reduce the effectiveness of these systems. In practice, rather than identifying a variety of disease classes it is often necessary to identify a retinal disease (present or absent) for initial and quick diagnosis. The binary classification will simply do this identification, simplifying the problem greatly and decreasing the computational complexity and model performance. Hence, the main goal of this study is to create a binary classification system to identify the retinal disease (disease vs normal) with the help of the RFMiD dataset. The suggested method is expected to achieve a better detection accuracy, better generalisation, low computational cost and develop a scalable and dependable system that would be able to provide early diagnosis as well as aid in the decision making processes of clinicians.

II. LITERATURE REVIEW

A. Description of relevant existing works

- **Automated Detection of Diabetic Retinopathy Using Custom Convolutional Neural Networks**

The paper mainly concentrates on a custom CNN network to detect retinal disease (mostly DR) with fundus images. The pre-processing phase is extremely low-level and most commonly, standard procedures such as resizing of images, normalization (scaling of pixels) and rudimentary augmentation is done. No sophisticated filtering methods (such as edge-enhancement or contrast-specific transforms) are highlighted, limiting enhancement of features at the input level. Training the model is realized with a relatively shallow custom CNN, which is trained on a Kaggle dataset, without pre-trained weights. The architecture comprises of a pooling, convolutional and fully connected layers. It can be easily trained, with binary classification and cross-entropy loss. The model has high accuracy 95 percent, which is attributed to the small dataset size, but probably due to bias in the dataset. The weaknesses of the paper are mainly its absence of solid pre-processing and inability to generalize. The model does not respond to intricate retinal differences because of the small size of the data and the binary characteristic of the data. Also, lack of methods to improve the features (such as vessel highlighting or contrast normalization) implies that the CNN has to learn all the information on the raw image, which decreases its efficiency. Another disadvantage of the model is that it is not flexible to multi-disease conditions and to clinical variability in real-life conditions [1].

- **:MGSCNN: Multi-Gradient Swarm Optimized CNN for Retinal Disease Classification**

The proposed hybrid model in this paper is a combination of CNN and Glowworm Swarm Optimization (GSO) to classify multi- retinal diseases. Simple normalization and dataset-specific processing take place in the pre-processing stage, and are not highly reliant on classical image processing filters. Rather, focus has been in optimizing feature selection and not improving input images. The training method is more sophisticated than the typical CNNs. CNN is a feature extractor and GSO algorithm is a feature parameter optimizer (weights or feature subsets). This hybrid training boosts the accuracy of classification (95) on the RFMiD data particularly in the multi-class problems. Nevertheless, optimization process is much more complex in terms of computational complexity and training time.

Efficiency and scalability is the most missing element. Despite its good performance, the model is computationally intensive and cannot be readily implemented into real-time systems. In addition, the lack of a good pre-processing means that the model is heavily dependent on optimization and not input quality. The procedure may not apply to other data sets since the procedure

involves parameter tuning. This approach is more than complex where simpler tasks are involved such as binary classification. [2]

- **Deep Learning-Based Multi-Layer CNN for Retinal Disease Classification Using RFMiD Dataset**

The paper will discuss deep CNN models (12-layer and 20-layer models) when classifying retinal diseases with the help of the RFMiD dataset. The pre-processing techniques are easy and they are resizing, normalization and restricted augmentation. There are no sophisticated filters (e.g. edge-detecting or intensity transforms) and thus the fine retinal features cannot be enhanced. The training plan is on enhancing the training depth to enhance feature extraction. The 20-layer CNN represents more complicated patterns as opposed to the 12-layer model. Nevertheless, more complex networks add overfitting particularly when there is a small amount of data. Optimized models have a better accuracy of approximately +89.5 per cent that is lower than the best performance achieved.

Overfitting and lack of generalization is the major gap. Lack of good pre-processing implies that the model can barely recognize minuscule structures such as micro-lesions. Besides, more complex architectures are introduced but not warranted to provide better performance. The model is not efficient as well and needs to be tuned. This paper points out that it is not necessary simply to add depth, which needs to be represented more accurately (pre-processing) and regularized.

- **Retinal Disease Classification Using CNN and Transfer Learning (VGG16)**

This paper compares a baseline CNN and a transfer learning model (VGG16) when it comes to classifying retinal diseases. The pre-processing stage includes resizing, normalization and possibly augmentation, but again does not include any complex image processing operations like contrast control or edge detection. It is clear that the educational approach can be enhanced using the transfer learning. The CNN model and the VGG16 have a high accuracy of approximately 83 percent and 90.8 percent respectively because the features are pretrained. The higher-layers can be fine-tuned and lower-layers frozen to perform training. This minimizes the time of training and enhances features extraction.

The loophole is on the disproportion of classes and insensitivity to diseases. Nevertheless, the model has low recall (0.65) of diseased cases, which is a missed significant positive samples. The model also remains premised on basic pre-processing which limits the capability to identify small retinal features. It is also binary in nature, and cannot be extended to cases of multi-diseases. Cost in terms of computation is moderate yet high as compared to basic CNN. [3]

- **Retinal Disease Detection Using EfficientNet and Hybrid Deep Learning Models,**

The advanced architectures such as EfficientNet, hybrid

models (DenseNet + MobileNet) related to the retinal disease classification are discussed in this paper. The pre-processing step consists of standard normalization and augmentation, but does not heavily incorporate classical image enhancement algorithms. The training plan employs the transfer learning based on the state-of-the-art architectures. EfficientNet is almost 93-94 percent accurate and hybrid models are almost 92.9 percent. They are trained with fine tuning and have an advantage when it comes to feature extraction that they have deeper and more optimized architectures.

But the difference is in computational complexity and difficulty in deployment. The models are also computationally expensive and therefore cannot be applied in low resource or real time applications. Furthermore, the lack of the highly developed pre-processing means that the models are fully leveraging deep learning to extract features, which may not be the best based on medical images. The other trade-off is the efficiency and accuracy. [4]

B. Different Deep Learning Methods for Retinal Image Analysis

TABLE I
COMPARISON OF DIFFERENT DEEP LEARNING METHODS FOR RETINAL IMAGE ANALYSIS

Method	Dataset	Accuracy	Efficiency	Limitation
Custom CNN	Kaggle DR	95.27%	High	Limited to binary classification, lacks generalization
MGSCNN (CNN + GSO)	RFMiD	95.09%	Medium	High computational complexity, difficult to deploy
Multi-layer CNN	RFMiD	89.59%	High	Overfitting due to deep architecture
VGG16 (Transfer Learning)	Kaggle	90.8%	Medium	Low recall due to class imbalance
EfficientNet / Hybrid	Fundus	92–93.8%	Low	Complex architecture, high resource requirement

C. Research Gaps of Existing Models

- **Deep Architectures Overfitting:** Deep CNN models (i.e. 20-layer networks) tend to overfit because of the small dataset size and insufficient regularization, which lowers their ability to perform on unseen data.
- **Problem of Class Imbalance:** The retinal datasets generally have limited diseased samples relative to normal samples which results to biased learning and reduced recollection of disease detection.
- **High Computational Cost:** More complex architectures and hybrid models (e.g., CNN with optimization methods) are more based on computing resources and are less suitable in real-time or resource-constrained settings.
- **Weak Generalization Ability:** Imbalance can hardly be completely addressed in pre-processing.

- **Overfitting Still Exists:** Models that are trained on particular datasets (such as RFMiD or Kaggle) do not in generalize to different datasets because of the differences in image quality, acquisition devices and population heterogeneity.
- **Low Multi-class Accuracy:** CNNs can be used effectively in binary classification and not in multi-disease classification, as there is overlap in features, and similarities across classes.
- **Lack of Pre-processing Integration:** Most studies have extensively used CNNs, but have not integrated low-level preprocessing methods, which results in the ineffective representation of features by raw images.

The use of Convolutional Neural Network (CNN)-based methods has greatly contributed to the detection of retinal diseases as it allows automatic feature extraction of fundus images. Many different architectures, including custom CNNs, VGG16, EfficientNet, and hybrid networks, are shown to be accurate on a wide range of datasets, such as Kaggle and RFMiD. Nevertheless, their performance is affected by such factors as network depth, preprocessing and quality of data. Although more accurate models are possible with deeper and optimized models, they can be overfitted, and may be expensive to compute. Also, issues like imbalance in classes and ineffective generalization remain. Generally, the CNN-based approaches have been effective although they need balanced design considerations to guarantee efficiency, robustness, and practicality of application. [1]–[5]

III. DATASET

A. RFMiD Dataset

The Retinal Fundus Multi-Disease Image Dataset (RFMiD) is a broad-based open-source dataset designed for automated retinal disease detection. Unlike earlier datasets focusing on limited conditions such as diabetic retinopathy or glaucoma, RFMiD includes a wide range of common and rare retinal pathologies, enabling more generalized AI models.

The dataset contains 3,200 retinal fundus images collected between 2009 and 2020. Images were captured using multiple fundus cameras, introducing variations in resolution and quality that reflect real-world clinical scenarios. Both high- and low-quality images are included, increasing the dataset's complexity and practical relevance. RFMiD supports multi-label annotations, allowing a single image to be associated with multiple diseases.

The dataset is divided into training (60%), validation (20%), and testing (20%) sets. Each image is linked to its corresponding labels via a CSV file. All annotations were verified by expert ophthalmologists to ensure high reliability [6].

RFMiD includes 46 classes (1 normal and 45 disease categories), covering conditions such as Diabetic Retinopathy (DR), Age-related Macular Degeneration (ARMD), Media Haze (MH), and others.

Despite its advantages, the dataset presents challenges such as class imbalance and lack of demographic information.

Evaluation_Set

Test_Set

Training_Set

ID	Disease_Type	DR	ARMD	MH	DN	MYA	BRVO	TSLN	ERM	LS	MS	CSR
1	1	1	1	0	0	0	0	0	1	0	0	0
2	1	1	1	0	0	0	0	1	0	0	0	0
3	1	1	1	0	0	0	0	0	0	0	0	0
4	1	1	0	0	0	0	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0	0	0	0	0
6	1	0	0	0	0	0	0	0	0	0	0	1
7	1	0	0	0	0	0	0	0	0	0	0	1
8	1	0	0	0	0	0	0	0	0	1	0	0
9	1	1	1	0	0	0	0	0	0	0	0	0
10	1	1	1	0	0	0	0	0	0	0	1	0

Fig. 1. Dataset Arrangement

Nevertheless, it remains a valuable resource for developing robust AI-based diagnostic systems [6].

B. RFMiD Dataset Details in Tabular Format

TABLE II
RFMiD DATASET DETAILS

Section	Details
Dataset Name	Retinal Fundus Multi-Disease Image Dataset (RFMiD)
Total Images	3200
No. of Subjects	Not mentioned
Dataset Type	Retinal Fundus Image Dataset
Task Type	Multi-label Classification
Data Split	Training: 60% (1920 images), Validation: 20% (640 images), Test: 20% (640 images)
Classes (All)	NL, DR, ARMD, MH, DN, MYA, BRVO, TSLN, ERM, LS, MS, CSR, ODC, CRVO, TV, AH, ODP, ODE, ST, AION, PT, RT, RS, CRS, EDN, RPEC, MHL, RP, CWS, CB, ODPM, PRH, MNF, HR, CRAO, TD, CME, PTCR, CF, VH, MCA, VS, BRAO, PLQ, HPED, CL
Subject Demography	Not mentioned
Age Range	Not mentioned
Annotation Process	Annotated by ophthalmologists and verified through expert consensus
Data Source Location	Eye clinic and research institute (India)
Image Acquisition	TOPCON 3D OCT-2000, Kowa VX-10, TOPCON TRC-NW300
Dataset Challenges	Highly imbalanced dataset
Use Case	AI-based retinal disease detection

C. Data Distribution

- Normal (NL): 669
- Diabetic Retinopathy (DR): 632
- Media Haze (MH): 523
- Optic Disc Cupping (ODC): 445
- Tessellation (TSLN): 304
- Drusen (DN): 230
- ARMD: 169
- Myopia (MYA): 167
- BRVO: 119
- Others: 1–10 images

IV. METHODOLOGY

The proposed algorithm combines low-level image processing with a high-level deep learning structure that refines the feature representation and boosts the performance of retinal disease classification. Raw images in medical imaging especially retinal fundus analysis commonly face issues of poor contrast, non-uniform lighting, noise, and poor view of important parts of the body. Such constraints can greatly affect the process of machine learning models in learning discriminative features. Hence, a powerful preprocessing pipeline is needed so as to clean up the input data prior to model training.

The first step in the procedure entails classical preprocessing methods to improve structural and intensity-related properties of the pictures. These functions underline the main diagnostic characteristics, such as blood vessels, edges of lesions, and minor abnormalities, with contrast amplification and edges sharpness. This step enhances better visual interpretability but it also enables better feature extraction in the learning process.

The resulting processed images are then given to the ResNet50 neural network, a deep convolutional network that learns features in a hierarchy. ResNet50 takes advantage of residual connections and overcomes the vanishing gradient issue, allowing deep models to be trained effectively. The network learns both low level and high level features, and finally classifies the binary (normal vs diseased) with better accuracy, generalization and clinical usability.

A. Low-Level Processing

Pre-processing plays a crucial role in improving image quality and highlighting discriminative features before feeding data into the neural network.

Sobel Filter

The Sobel filter is a first-order derivative operator used for edge detection by computing intensity gradients in horizontal and vertical directions Fig. 3. Mathematically, it applies convolution kernels:

Horizontal Kernel:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Vertical Kernel:

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

Gradient Magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

Purpose in Retinal Images:

- Enhances blood vessels and lesion boundaries
- Extracts structural information

Advantage:

- Simple and computationally efficient
- Highlights important edge features [7], [8].

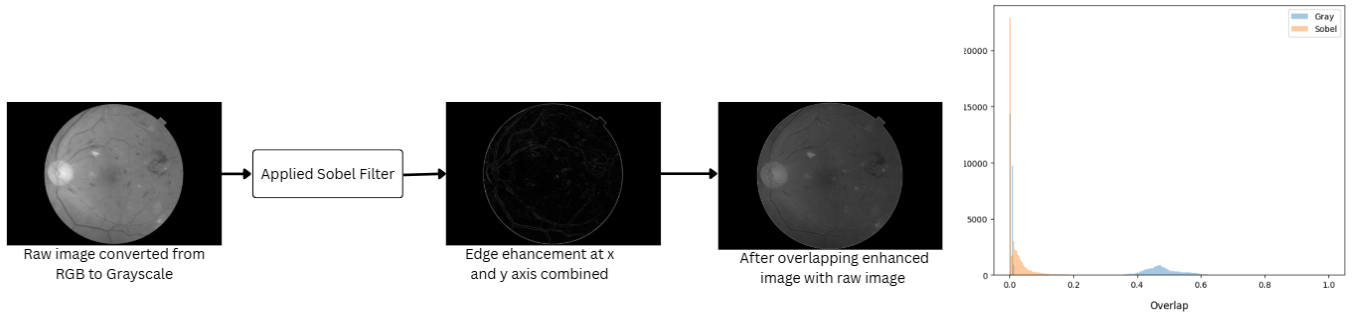


Fig. 2. Sobel Filter Application and Histogram

B. High-Level Processing

ResNet50 is based on the concepts of a Convolutional Neural Network (CNN) and a Neural Network.

Convolutional Neural Network (CNN):

A CNN is a deep learning architecture designed for image processing tasks. It consists of:

- **Convolutional layers** → Feature extraction
- **Pooling layers** → Dimensionality reduction
- **Fully connected layers** → Classification

Neural Network:

A Neural Network is a computational model inspired by the human brain, consisting of interconnected nodes (neurons) that learn patterns from data. [10].

ResNet50 ResNet50 is a 50-layer deep residual network, which adds skip (residual) connections to overcome the vanishing gradient problem. Fig. 6 Instead of learning direct mapping, it learns: [13].

$$F(x) = H(x) - x$$

Purpose in This Work

- Conducts a binary classification (disease vs normal).
- Descends hierarchical features of already processed pictures.

Advantages of ResNet50

- Avoids vanishing gradient problem.
- Facilitates very deep architecture.
- Enhances accuracy and convergence
- Efficient feature learning

The fundamental concept of residual learning is expressed as:

$$y = F(x) + x$$

where $F(x)$ represents the residual function and x is the input. This structure allows gradients to flow directly through the network, ensuring stable and efficient training even in very deep architectures. Fig. 3

C. Training Strategy

This study training plan will build up on the training of the deep learning architectures to be effective in the classification of retinal disease in various conditions systematically. Firstly, a default Convolutional Neural Network (CNN) model will

be created and trained with the use of the untouched RFMiD data set without any preprocesses or optimization methods. This method is aimed at creating a baseline performance, and to monitor the behavior of a handcrafted CNN when directly subjected to raw retinal fundus images. CNNs are known to be highly adaptable because of their ability to learn hierarchical features of image data automatically, which is why they are mostly applicable in medical image classification [9]. Nonetheless, earlier research has demonstrated that models that are trained with raw retina images can experience poorer performance because of noise, low contrast, and light imbalances that are inherent to fundus imaging [11].

After this, the transfer learning strategy will be adopted with ResNet-50 architecture. The deep convolutional neural network ResNet-50 is a 50-layer convolutional neural network that uses residual (skip) connections that enable the network to learn more deeply resolved representations and counter the vanishing gradient problem [3]. During this stage, the model will be trained on the same raw data as well, so that it can be compared to the custom CNN fairly. Earlier studies have shown that ResNet based models usually perform better than shallow CNN based architecture with respect to retinal disease classification because it can embed complicated feature representations [18]. This analogy will give an idea of the benefits of deep residual learning compared to the traditional CNN design in the absence of preprocessing.

The second step will include preprocessing by using the Sobel filter to update edge information in the retinal images. Edge-based filters (Sobel) are well known in order to outline structural features including blood vessels and lesion edges that are important in the accurate detection of disease [19]. Research has revealed that the combination of edge-enhancement methods and CNN models can greatly enhance the accuracy of classification by highlighting diagnostically important features [11]. Also, preprocessing methods have been extensively documented to improve the performance of the models used in the analysis of retinal image by increasing image quality and visibility of features [20].

Lastly, where needed, post-processing methods will be

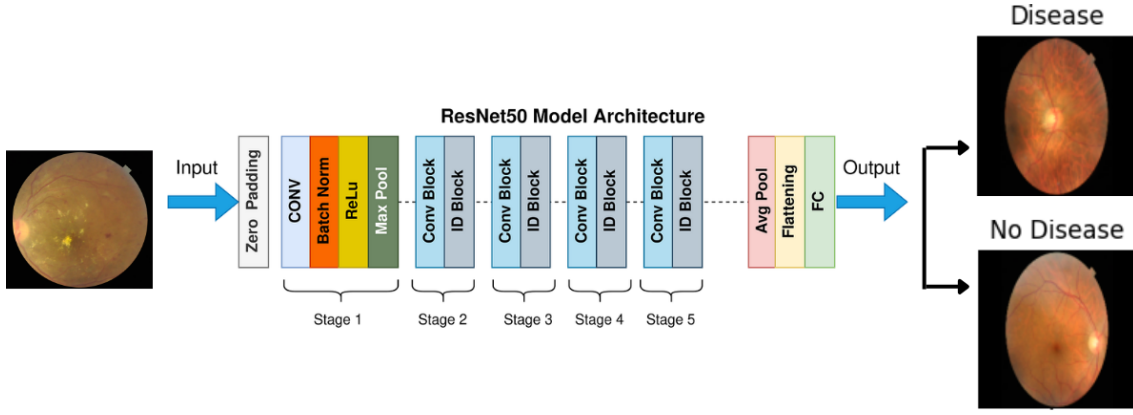


Fig. 3. Resnet 50 Architecture [12]

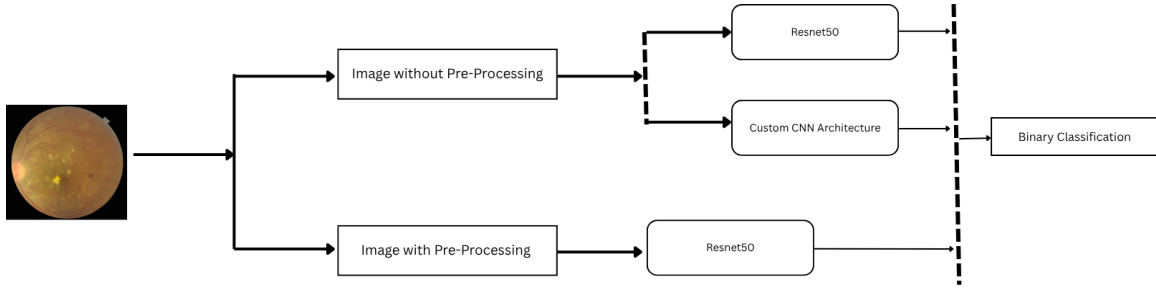


Fig. 4. Summarized Version of our Training Strategy

considered to further optimize model predictions and enhance overall measures of performance including the accuracy, precision, recall, and F1-score. All the models will be evaluated based on the standard performance measures and confusion matrices to provide a complete comparison. The multi-stage training approach will allow carrying out a comprehensive exploration of the effects of preprocessing and model complexity on classification performance. This paper will compare a custom CNN, a deep residual network, and a preprocessing-enhanced ResNet-50 model, and determine which approach is best to use in reliable and accurate retinal disease detection [20].

D. Evaluation Metrics

The performance of the classification model is evaluated using standard metrics derived from the confusion matrix shown on Fig. 5. [16]

A **confusion matrix** is a utility which is used to assess the performance of classification in terms of predicted and actual values. It is comprised of four elements namely True Positive, True Negative, False Positive and False Negative. It assists in detecting various classes of errors and offers greater insight than merely accuracy particularly in medical diagnosis so as to reduce the critical misclassifications. [14], [15]

- **Accuracy:** Measures the overall correctness of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** Indicates the proportion of correctly predicted positive cases among all predicted positives.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- **Recall (Sensitivity):** Measures the ability of the model to correctly identify actual positive cases.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

- **Specificity:** Evaluates the model's ability to correctly identify negative cases.

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

- **F1-Score:** Represents the harmonic mean of Precision and Recall, balancing both metrics.

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative.

V. RESULTS AND FINDINGS

A. ResNet-50 without pre-processing

The ResNet50 model trained on the complete dataset, without any low-level preprocessing indicates that the model is correctly identifying 134 non-disease cases (TN) and is not

identifying any disease cases ($TP = 0$) leading to 506 false negatives and zero false positives. This is also evident in the evaluation metrics with the recall (0) and F1 score (0) being 100 percent failure to detect disease with the precision (0) also giving confirmation that no positive prediction was made. The specificity is 100, that is, the model perfectly classifies healthy samples, but the overall accuracy is very low (20.9percent), which indicates that the performance of the model is strongly biased towards the negative class.

This finding justifies that the model is grossly underfitting and has learnt a trivial decision boundary by classifying all samples as non-disease. This is probably because of poor extracting of features of raw images, the lack of fine-tuning of the pretrained ResNet layers, and the probability of data imbalance leading to the model favoring the majority or the simpler class. The reported value of AUC indicates that there is good separability, but the selected classification threshold does not allow the model to recognize positive cases. To resolve this problem additional fine-tuning of the pretrained network is necessary, as well as class balancing methods (weighted loss or resampling). Further, it can be used to adjust the decision threshold and use suitable preprocessing or data augmentation to ensure that the model learns more discriminative features to better detect the disease.

The pretrained CNN model's ability to identify retinal diseases is limited, according to the results obtained. While the model accurately categorizes normal images, it is unable to recognize cases of illness. As a result, the F1-score and recall are both zero, indicating poor disease detection performance. Inadequate feature extraction, insufficient fine-tuning of pre-trained layers, or dataset imbalance could all contribute to this limitation. Future research will concentrate on enhancing the performance through deeper fine-tuning of pretrained CNN architectures, class balancing strategies, and improved data augmentation. Fig. 5

B. ResNet-50 With Sobel Filter

The results of the ResNet50 model that is trained on the entire dataset using Sobel-filtered images indicate a very skewed classification performance. The confusion matrix indicates that the model makes many false negatives (245) than false positives (9) indicating that the model is often not able to identify diseased cases but it is highly conservative in predicting disease. It is also shown in the metrics of evaluation: the model has a very high accuracy (0.9667) and specificity (0.9328), i.e. when the model predicts a disease the prediction is almost always correct and the false prediction is not made very often, but the recall (0.5158) is very low. This implies that it is missing close to half of the true diseased cases, which is a severe limitation in medical diagnosis tasks where it is more significant to find the positive cases, as opposed to avoiding false alarms. The fact that the F1 score (0.6727) and the overall accuracy (0.6031) are relatively moderate, and that the model is biased towards the negative class also confirms that the model is biased. This act could be ascribed to various factors. First, edge-enhancing filters such as the Sobel filter

can suppress valuable texture and color-based information which would be important in the detection of subtle retinal abnormalities, hence resulting in lower sensitivity. Second, the model seems to prefer the majority or more easily distinguishable class which might be because of the imbalance among classes or because of the nature of features that are learned by edge-enhanced images. Also, ResNet50, with frozen layers or post-training fine-tuning, might not learn the domain-specific attributes of medical images, particularly when preprocessing distorts the underlying feature distribution. In order to overcome these problems some improvements can be taken into consideration. To make the model more sensitive to diseased cases, it is necessary to change the decision threshold (e.g. make it smaller than 0.5), penalize false negatives more (class weighting), or it can also be done with loss functions like focal loss. Moreover, rather than using Sobel filtering exclusively, raw images could be used together with edge-enhanced images or less aggressive preprocessing techniques might be used to preserve more diagnostic details. Further layers of ResNet50 can also be fine-tuned instead of being frozen which can enhance feature adaptation. On the whole, the model is highly accurate and specific, but its low recall reveals a major drawback of this model in terms of disease detection, which may require changes and focus more on the sensitivity in this medical classification task.

This model shows that it underfits with a distinct bias towards the negative class as shown by low recall (0.5158) and high false negative (245) although the precision is high (0.9667). It implies that the model is sure of its prediction ability of disease, but it does not reflect a lot of the real disease cases. This is because the Sobel filter mainly amplifies edge information and inhibits the valuable texture and intensity features which are important in identification of small retinal abnormalities. Also, the fact that the ResNet architecture can be easily adapted to the needs of various domains can be the cause of such behavior. The finer-tuning of deeper layers of the model, decreasing reliance on Sobel filtering or using it together with raw images and sensitivity through class weighting or threshold adjustment can be used to improve performance. Fig. 6

C. Classification based on Custom CNN

Using the custom CNN model demonstrates that the model accurately detects 505 cases of disease (TP) and just one case of non-disease (TN) but yields a very large number of false positives (133) and an almost zero number of false negatives (1). This trend is expressed in the evaluation measures: the model has a perfect recall (1.0000), meaning that it identifies almost all cases of disease, and a fairly high F1 score (0.8831), as well as a mediocre precision (0.7906). The specificity is however very low (0.0000) indicating that the model is not able to accurately detect healthy samples and defines nearly all as diseased. The accuracy (0.7906) is not very low but is deceptive because the model has a strong inclination towards a positive class. Fig. 7

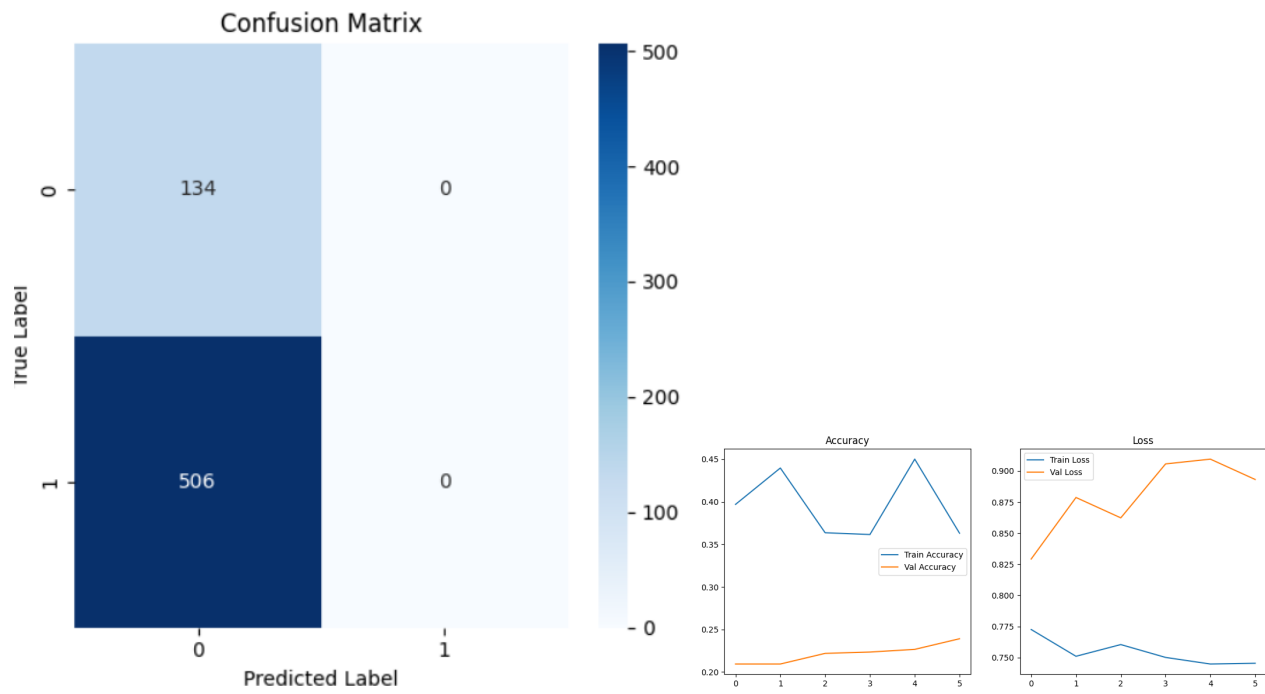


Fig. 5. Confusion Matrix and Evaluation of ResNet-50

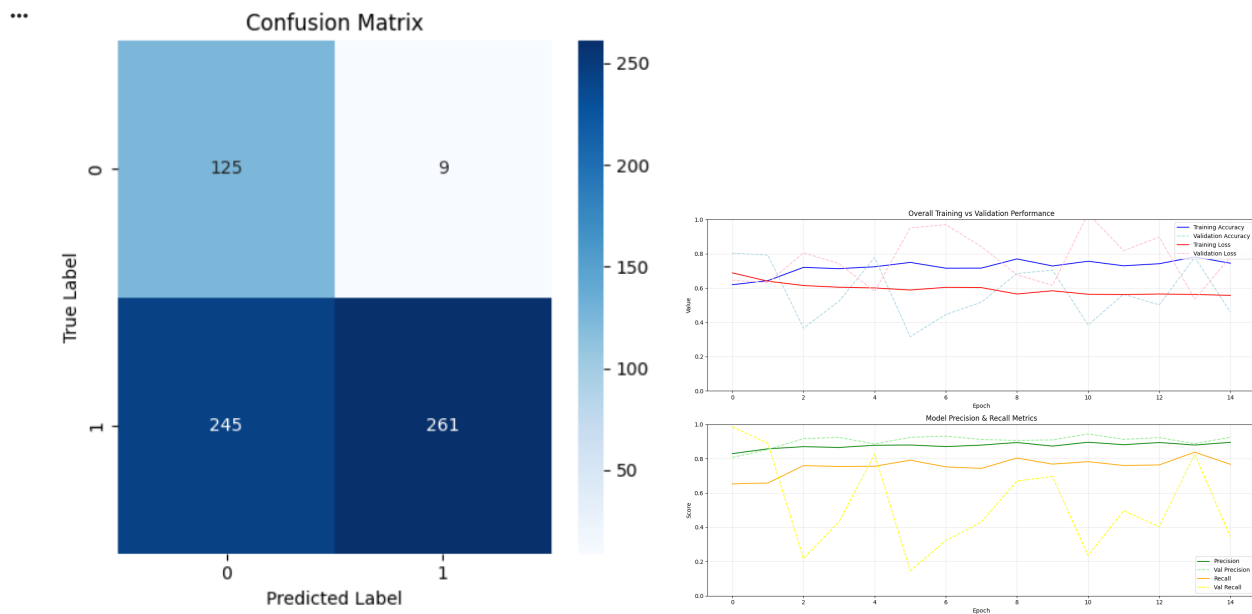


Fig. 6. Confusion Matrix and Evaluation of Sobel Filter

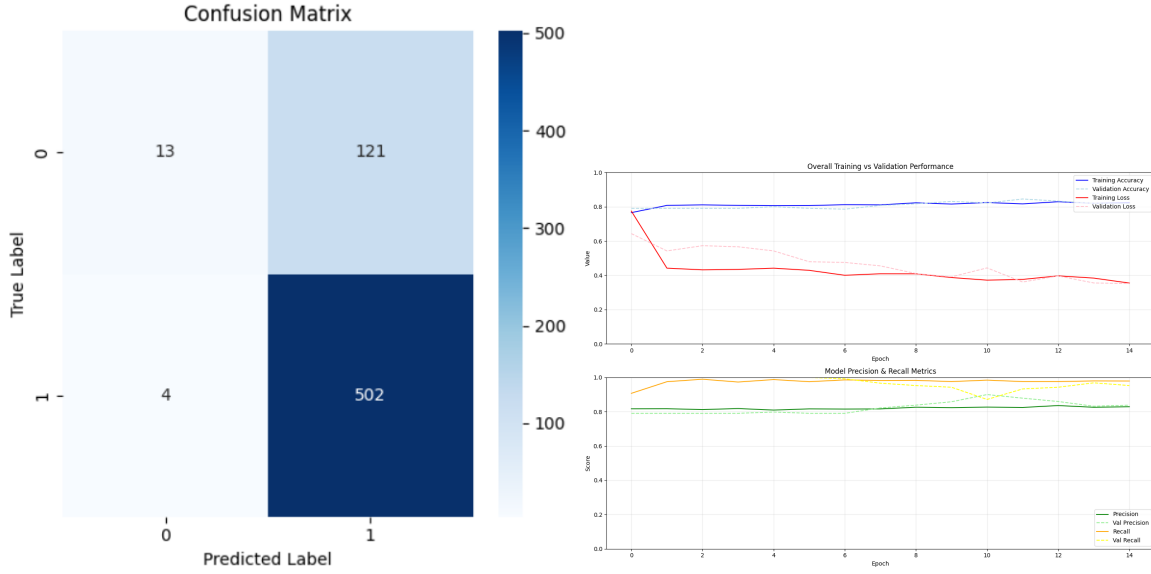


Fig. 7. Confusion Matrix and Evaluation of Custom CNN

This finding justifies that the behavior of CNN is highly biased on making predictions of the type of disease like the one seen in certain ResNet design which is similar to the behavior of a near-trivial classifier. This is probably because of the imbalance of classes in the data, with the model being biased in reducing the loss by classifying to the prevalent class, and inadequate learning of the discriminative features of the non-disease group. Moreover, Flatten and dense layers can lead to overfitting of the prevalent patterns in the model without refining the subtle differences of the healthy images. To solve this problem, methods of class balancing, including weighted loss functions, resampling, and changing the decision threshold to minimize false positives, should be used. The model can also include more effective feature learning techniques, like increased convolutional depth or mixed with feature-enhancing preprocessing techniques, which can make the model have a more balanced sensitivity and specificity.

TABLE III
COMPARISON OF DIFFERENT ARCHITECTURE

Metric	Sobel (ResNet)	Custom CNN	No Preprocessing (ResNet)
Accuracy	0.603	0.791	0.209
Precision	0.967	0.791	0.000
Recall	0.516	1.000	0.000
Specificity	0.933	0.000	1.000
F1 Score	0.673	0.883	0.000
Issue Type	Underfitting	Overfitting (Bias)	Severe Underfitting
Key Problem	Misses disease	Predicts all disease	No disease detection

VI. CONCLUSION

This study explored the performance of different deep learning architectures for binary classification of retinal diseases using the RFMiD dataset [6]. Three main methods were compared: a ResNet50 model, which lacked preprocessing, a Sobel filter-enhanced ResNet50 model, and a custom CNN architecture. The findings reveal that both models have their

own unique advantages and drawbacks, mostly depending on the preprocessing methods, the complexity of the architecture and the nature of the data. The ResNet50 model that was trained on raw images demonstrated strong underfitting, was unable to identify diseased individuals, and made very biased predictions in the normal category. This points out the weaknesses of directly applying deep architectures without adequate feature augmentation or fine-tuning. The Sobel filter-enhanced ResNet50 model enhanced the general accuracy and specificity, meaning that it detected more normal samples. But it has a relatively low recall implying that a considerable number of diseased instances were overlooked and therefore it is not suitable in medical applications where sensitivity is paramount [7].

On the other hand, the custom CNN model had a flawless recall, being able to detect almost all diseased cases. It, however, had low specificity and the vast majority of normal samples were classified as diseased. This indicates overfitting and high positive bias, which is probably due to imbalance between the class and lack of feature differentiation [13].

On the whole, the results show that none of the models could be considered to provide the best balance between sensitivity and specificity. It is pointed out in the study that a good retinal disease classification must be based on a combination of powerful preprocessing, a proper model selection, and class balancing strategies as well as prudent decision threshold tuning. The main direction of future work should be to enhance the generalization, combine hybrid preprocessing methods, and use sophisticated loss functions to diminish the bias. These are some improvements that can be made to develop accurate, reliable and scalable automated systems in detecting early retinal diseases in clinical settings.

VII. CONTRIBUTION

The authors also acknowledge the contributions of each team member. Mst. Shikha Khatun used ResNet50 in the pretrained model. Raisa Tafannum came up with the Custom Convolutional Neural Network(CNN)architecture. Jugumaya Saha Disha did her part by introducing ResNet50 with low-level processing methods.The joint work of all participants contributed greatly to the success of this work.

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